

Inter-VLAN Routing

Part A: Theory

Methods of Inter-VLAN Routing

VLANs segment a switched network into separate broadcast domains, which improves security and reduces unnecessary traffic. However, devices in different VLANs cannot communicate with each other without a Layer 3 device to route between them. Two common methods achieve this: Router-on-a-Stick and Layer 3 switching.

1. Router-on-a-Stick

In this method, a single physical router interface is connected to a trunk port on the switch. The router interface is divided into logical sub-interfaces, one per VLAN, each configured with 802.1Q encapsulation and an IP address that serves as the default gateway for that VLAN. All inter-VLAN traffic passes through this one physical link.

2. Layer 3 Switch (Switched Virtual Interfaces)

A Layer 3 switch performs routing internally using Switched Virtual Interfaces (SVIs) — one logical “interface vlan X” per VLAN. Because routing happens in switch hardware (ASICs) rather than over a single trunk link, this method is faster and avoids the single-link bottleneck of router-on-a-stick.

Task 1: In Cisco Packet Tracer, create three VLANs named Foodies, Shoppers, and Streamers to represent users of Zomato, Flipkart, and Spotify, and assign at least one PC to each VLAN.

Three VLANs were created on the Multilayer Switch to represent three different user groups, and one PC was assigned to each VLAN as an access port.

```
Switch(config)# vlan 10
Switch(config-vlan)# name Foodies
Switch(config-vlan)# exit
Switch(config)# vlan 20
Switch(config-vlan)# name Shoppers
Switch(config-vlan)# exit
Switch(config)# vlan 30
Switch(config-vlan)# name Streamers
Switch(config-vlan)# exit
```

PC	VLAN	Represents	Port	Mode
PC0	10 – Foodies	Zomato	Fa0/1	Access
PC1	20 – Shoppers	Flipkart	Fa0/6	Access
PC2	30 – Streamers	Spotify	Fa0/11	Access

Network topology showing Router0 (2620XM), the multilayer switch, and the three PCs each connected to their respective VLAN access ports:

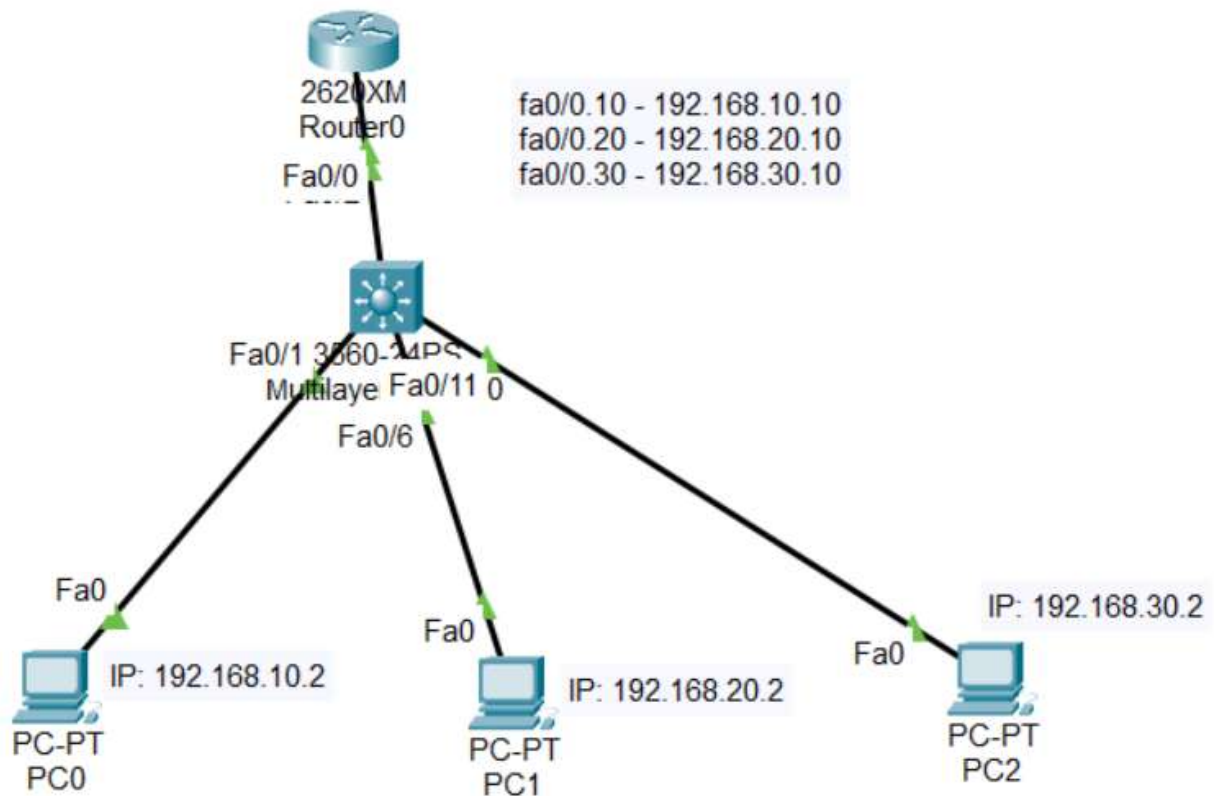


Figure 1: Network topology — Router0, Multilayer Switch0, and PC0/PC1/PC2

Task 2: Configure a router-on-a-stick setup in Packet Tracer so that PCs in the Foodies, Shoppers, and Streamers VLANs can all ping each other using sub-interfaces on the router (e.g., GigabitEthernet0/0.10, .20, .30).

The switch port facing the router (Fa0/24) was configured as a trunk carrying all three VLANs. On the router, three sub-interfaces were created off the physical Fa0/0 interface, each with 802.1Q encapsulation and an IP address acting as that VLAN's default gateway.

Switch – Trunk Port Configuration

```
Switch(config)# interface fastEthernet 0/24
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk allowed vlan 10,20,30
```

Router – Sub-interface Configuration

```
Router(config)# interface fastEthernet 0/0
Router(config-if)# no shutdown
Router(config-if)# exit

Router(config)# interface fastEthernet 0/0.10
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 192.168.10.10 255.255.255.0
Router(config-subif)# exit

Router(config)# interface fastEthernet 0/0.20
Router(config-subif)# encapsulation dot1Q 20
Router(config-subif)# ip address 192.168.20.10 255.255.255.0
Router(config-subif)# exit
```

```
Router(config)# interface fastEthernet 0/0.30
Router(config-subif)# encapsulation dot1Q 30
Router(config-subif)# ip address 192.168.30.10 255.255.255.0
Router(config-subif)# exit
```

Router0's console log confirming Fa0/0 and all three sub-interfaces (0.10, 0.20, 0.30) came up successfully, along with CDP neighbor output showing the switch connections on Fa0/2 and Fa0/24:

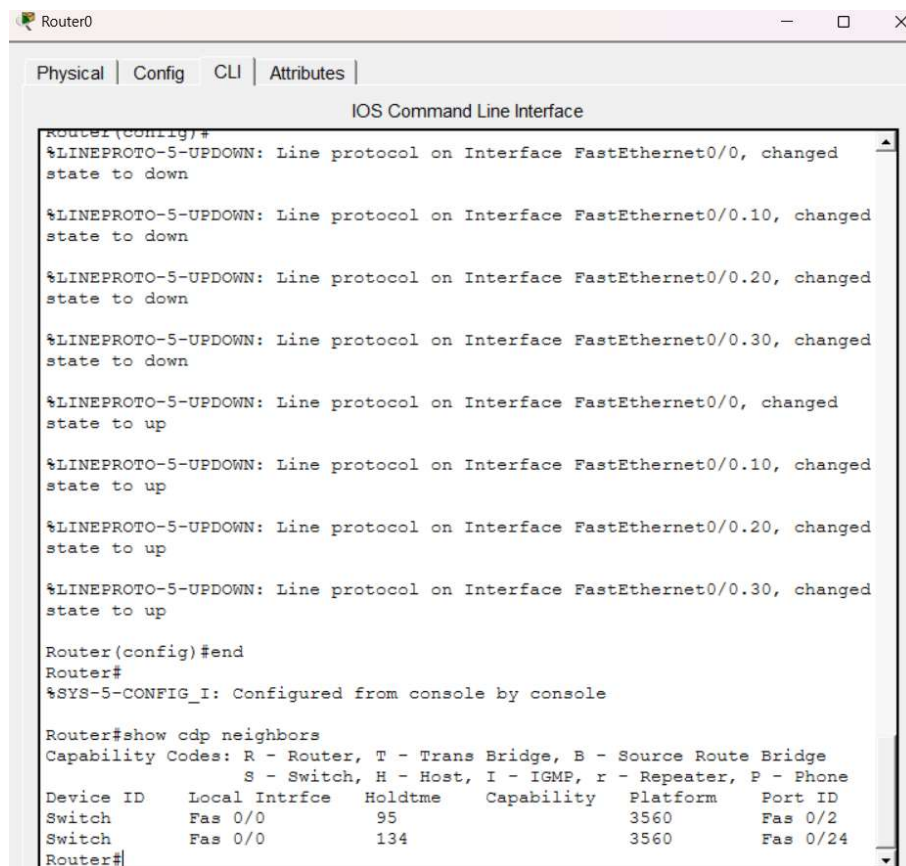


Figure 2: Router0 interface up/down log and show cdp neighbors output

Task 3: Assign IP addresses to each PC in your VLANs and set their default gateways to the corresponding router sub-interface IPs to enable inter-VLAN communication.

Each PC was assigned a static IP address within its VLAN's subnet, with the default gateway set to the corresponding router sub-interface IP.

PC	VLAN	IP Address	Subnet Mask	Default Gateway
PC0 (Zomato)	10	192.168.10.2	255.255.255.0	192.168.10.10
PC1 (Flipkart)	20	192.168.20.2	255.255.255.0	192.168.20.10
PC2 (Spotify)	30	192.168.30.2	255.255.255.0	192.168.30.10

Task 4: Simulate a scenario where users in the Shoppers VLAN (Flipkart) cannot ping users in the Streamers VLAN (Spotify). Identify and fix the misconfiguration causing the issue, then verify connectivity. Hint: Check VLAN assignments, router sub-interface configurations, and trunk port settings.

Symptom

PC0 (VLAN 10, Foodies) was unable to reach PC1 in VLAN 20 (Shoppers). A ping from PC0 to 192.168.20.2 resulted in 100% packet loss:

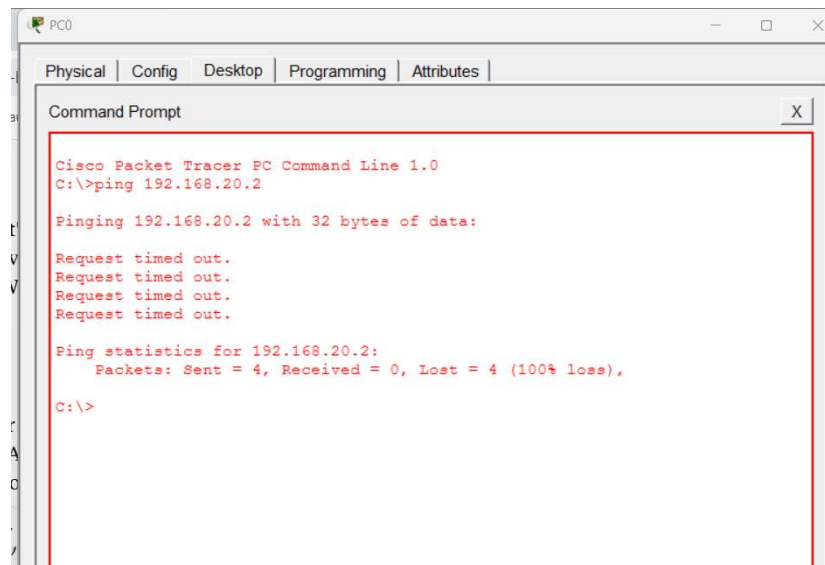
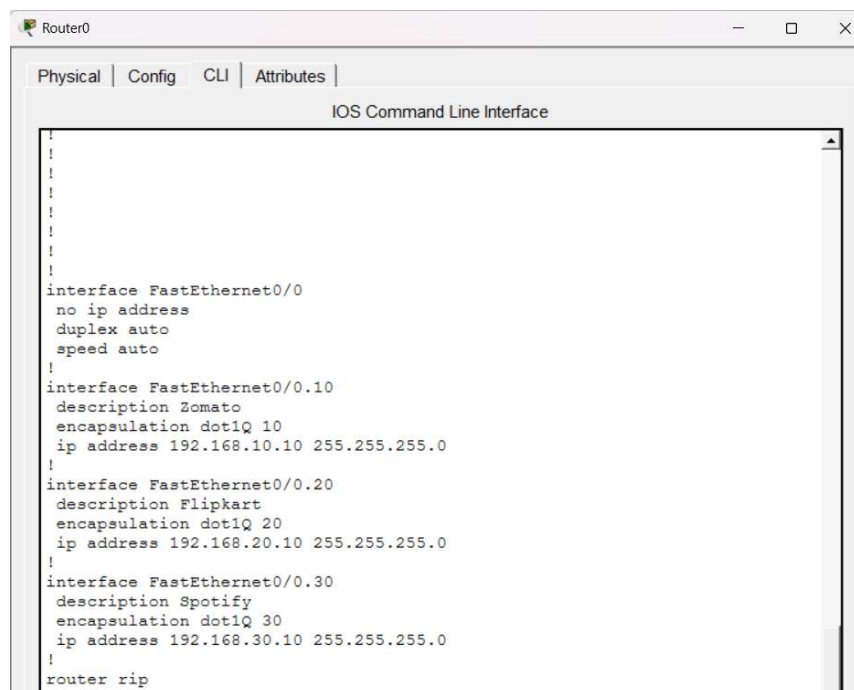


Figure 3: Failed ping from PC0 to 192.168.20.2 — 4 sent, 0 received, 100% loss

Diagnosis

The router's Fa0/0 physical interface and all three sub-interfaces (0.10, 0.20, 0.30) were confirmed up via the router console log, ruling out a router-side interface problem. Attention then turned to the switch.



Running show vlan brief on the multilayer switch showed VLANs 10, 20, and 30 correctly created with the expected access ports assigned and show interfaces trunk was then run to confirm Fa0/24 was correctly trunking all three VLANs:

```
Physical | Config | CLI | Attributes |
IOS
allowed      Set allowed VLAN characteristics when interface is in tru
              mode
encapsulation Set trunking encapsulation when interface is in trunking
native       Set trunking native characteristics when interface is in
              trunking mode
Switch(config-if)#switchport trunk allowed ?
vlan         Set allowed VLANs when interface is in trunking mode
Switch(config-if)#switchport trunk allowed vlan ?
WORD         VLAN IDs of the allowed VLANs when this port is in trunking mode
add          add VLANs to the current list
all          all VLANs
except       all VLANs except the following
none         no VLANs
remove       remove VLANs from the current list
Switch(config-if)#switchport trunk allowed vlan 10,20,30
Switch(config-if)#do wr
Building configuration...
[OK]
Switch(config-if)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/24    on        802.1q         trunking    1

Port      Vlans allowed on trunk
Fa0/24    10,20,30

Port      Vlans allowed and active in management domain
Fa0/24    10,20,30

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/24    10,20,30

Switch#
```

Figure 5: show interfaces trunk confirming Fa0/24 is trunking, VLANs 10,20,30 allowed and active

Verification

With the trunk correctly configured, PC1 was used to ping all three router gateway sub-interfaces. Every ping succeeded with 0% packet loss, confirming full inter-VLAN connectivity was restored:

```
PC1
Physical | Config | Desktop | Programming | Attributes |
Command Prompt

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.20.10: bytes=32 time<1ms TTL=255
Reply from 192.168.20.10: bytes=32 time<1ms TTL=255
Reply from 192.168.20.10: bytes=32 time<1ms TTL=255
Reply from 192.168.20.10: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time<1ms TTL=255
Reply from 192.168.10.10: bytes=32 time<1ms TTL=255
Reply from 192.168.10.10: bytes=32 time<1ms TTL=255
Reply from 192.168.10.10: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:

Reply from 192.168.30.10: bytes=32 time<1ms TTL=255
Reply from 192.168.30.10: bytes=32 time<1ms TTL=255
Reply from 192.168.30.10: bytes=32 time<1ms TTL=255
Reply from 192.168.30.10: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
```

Figure 6: Successful pings from PC1 to 192.168.20.10, 192.168.10.10, and 192.168.30.10 — 0% loss

Task 5: Compare the router-on-a-stick method you just configured with inter-VLAN routing using a Layer 3 switch: list two advantages and two disadvantages of each approach in a table.

	Router-on-a-Stick	Layer 3 Switch (SVI)
Advantage 1	No need for expensive Layer 3–capable switch hardware	Routing performed in switch hardware (ASIC) — significantly faster
Advantage 2	Simple to set up using an existing router	No single trunk link bottleneck between VLANs
Disadvantage 1	All inter-VLAN traffic is bottlenecked through one trunk link	Higher hardware cost (Layer 3 switch required)
Disadvantage 2	Router becomes a single point of failure and added latency for inter-VLAN traffic	Slightly more complex initial setup (SVIs plus enabling IP routing)